Practice 1: Metric length

Name: ____________________ Date: __________

Use the metric relationships. Show your work and include units.

1. Convert 6 meters to centimeters.

Work space

2. Convert $4{,}300$ millimeters to meters.

Work space

Practice 2: Metric mass

Name: ____________________ Date: __________

Remember: 1 kilogram $= 1{,}000$ grams and 1 gram $= 1{,}000$ milligrams.

1. Convert 7 kilograms to grams.

Work space

2. Convert $5{,}600$ grams to kilograms.

Work space

Practice 3: Metric capacity

Name: ____________________ Date: __________

Use 1 liter $= 1{,}000$ milliliters.

1. Convert 9 liters to milliliters.

Work space

2. Convert $2{,}750$ milliliters to liters and milliliters.

Work space

Practice 4: Customary length

Name: ____________________ Date: __________

Use 12 inches $= 1$ foot, 3 feet $= 1$ yard, and $5{,}280$ feet $= 1$ mile.

1. Convert 7 feet to inches.

Work space

2. Convert 5 yards to feet.

Work space

Practice 5: Customary mass and capacity

Name: ____________________ Date: __________

Use 16 ounces $= 1$ pound, 2 cups $= 1$ pint, 2 pints $= 1$ quart, and 4 quarts $= 1$ gallon.

1. Convert 6 pounds to ounces.

Work space

2. Convert 3 gallons to quarts.

Work space

Practice 6: Converting mixed units

Name: ____________________ Date: __________

Rewrite each measurement in the requested single unit.

1. Convert 4 feet 7 inches to inches.

Work space

2. Convert 3 yards 2 feet to feet.

Work space

Practice 7: Decimal conversions

Name: ____________________ Date: __________

Use place value to convert. Write a decimal when it makes sense.

1. Convert 3.6 meters to centimeters.

Work space

2. Convert 475 centimeters to meters.

Work space

Practice 8: Choosing a useful unit

Name: ____________________ Date: __________

Choose the most reasonable unit, then explain briefly.

1. Would you measure the length of a pencil in inches, feet, or miles? Explain.

Work space

2. Would you measure the amount of water in a bathtub in milliliters, liters, or kiloliters? Explain.

Work space

Practice 9: Comparing measurements

Name: ____________________ Date: __________

Convert to the same unit before comparing. Use $>$, $<$, or $=$.

1. 3 meters ___ 280 centimeters

Work space

2. 2 kilograms ___ $1{,}950$ grams

Work space

Practice 10: Adding and subtracting measurements

Name: ____________________ Date: __________

Line up units or convert first.

1. Find 2 m 45 cm $+1$ m 80 cm. Give the answer in meters and centimeters.

Work space

2. Find 5 kg 200 g -2 kg 750 g. Give the answer in grams.

Work space

Practice 11: Multiplying and dividing measurements

Name: ____________________ Date: __________

Solve and label each answer.

1. Four boards are each 1.25 meters long. What is their total length in meters?

Work space

2. A jug contains 6 liters of water. All the water is shared equally among 8 empty bottles. How many liters go in each bottle?

Work space

Practice 12: Time-unit conversions

Name: ____________________ Date: __________

Use 60 seconds $= 1$ minute, 60 minutes $= 1$ hour, and 24 hours $= 1$ day.

1. Convert 7 minutes to seconds.

Work space

2. Convert 3 hours 25 minutes to minutes.

Work space

Practice 13: Elapsed time

Name: ____________________ Date: __________

Find how much time passes. Show a timeline or count by hours and minutes.

1. A movie starts at 6:35 p.m. and ends at 8:20 p.m. How long is it?

Work space

2. Practice begins at 3:48 p.m. and lasts 1 hour 27 minutes. When does it end?

Work space

Practice 14: Perimeter and measurement

Name: ____________________ Date: __________

Find the perimeter. Remember to use the same unit on every side.

1. A rectangle is 8 cm long and 5 cm wide. Find its perimeter.

Work space

2. A garden has sides 4 m, 7 m, 4 m, and 7 m. Find its perimeter.

Work space

Practice 15: Area and square units

Name: ____________________ Date: __________

Area measures the space inside a rectangle. Use $A = length \times width$.

1. Find the area of a rectangle 9 cm by 4 cm.

Work space

2. A floor is 12 ft long and 8 ft wide. How many square feet does it cover?

Work space

Practice 16: Volume and cubic units

Name: ____________________ Date: __________

For a rectangular prism, use $V = length \times width \times height$.

1. Find the volume of a box 5 cm by 3 cm by 2 cm.

Work space

2. A fish tank is 10 in long, 4 in wide, and 6 in high. Find its volume.

Work space

Practice 17: Temperature

Name: ____________________ Date: __________

Find the temperature change or new temperature. Include degrees Celsius in your answers.

1. The temperature is $18°$C in the morning and $25°$C in the afternoon. How many degrees did it rise?

Work space

2. A greenhouse is $12°$C. It warms by $7°$C. What is the new temperature?

Work space

Practice 18: Multi-step measurement problems

Name: ____________________ Date: __________

Plan the unit conversions before calculating.

1. A runner completes 3 laps of a 400-meter track and then runs 250 meters. How many kilometers does the runner travel?

Work space

2. A baker has 5 kg of flour and uses 750 g. How many grams remain?

Work space

Practice 19: Measurement data

Name: ____________________ Date: __________

Use the data to answer each question. Show calculations.

1. Three packages weigh 1.2 kg, 850 g, and 2 kg. What is their total mass in grams?

Work space

2. Four students measure a table: 120 cm, 1.35 m, 118 cm, and 1.22 m. Which measurement is longest? Give it in centimeters.

Work space

Practice 20: Mixed review

Name: ____________________ Date: __________

Solve each problem. Check that your final unit answers the question.

1. Convert 4.75 meters to centimeters.

Work space

2. A hike begins at 8:40 a.m. and ends at 11:15 a.m. How long is the hike?

Work space